

WASP-43b

R-0614

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180307 · created 2026-07-25T09:43:54+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458184.85966 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.013
Transit depth (Rp/Rs)^2	3.1 +/- 0.46 [%]
Orbital Inclination (inc)	81.07 +/- 0.69
Airmass coefficient 1 (a1)	0.969 +/- 0.01
Airmass coefficient 2 (a2)	0.0174 +/- 0.003
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	#1 - [328, 95]
Optimal Method	PSF photometry
Transit Duration (day)	0.0485 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.176 ± 0.013 vs 0.1594 (deviation 1.28σ)
Duration fit vs published	0.0485 d vs 0.0512 d (deviation 1.02σ)
Mid-time O–C	-2.7 min
Depth SNR	13.5
Residual scatter	1.06 %

Light curve

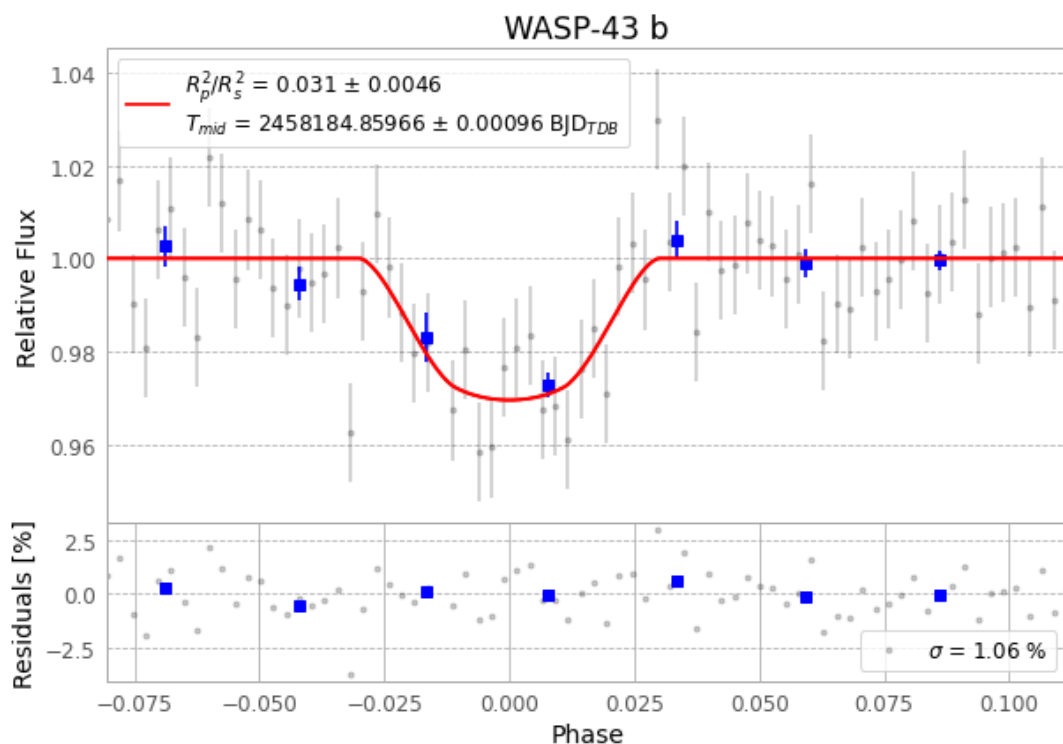


Figure 1 — FinalLightCurve_WASP-43 b_2018-03-06.png (SHA-256 4fa4b23ba09ded6e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 290], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2ecd4055f895bfba...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

76 FITS frames (50 MB), WASP-43180307065412.FITS → WASP-43180307103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180307.log (d5c592417210...), FinalLightCurve_WASP-43 b_2018-03-06.png (4fa4b23ba09d...), FinalLightCurve_WASP-43 b_2018-03-06.csv (4f4fe3a4acb9...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 09:43Z.